

Student Number: _____

Student Name _____

Teacher Name: _____



ABBOTSLEIGH

AUGUST 2011

YEAR 12

ASSESSMENT 4

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 120

- Attempt Questions 1-8.
- All questions are of equal value.
- Answer each question in a new booklet.

Outcomes assessed

HSC course

- E1** appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems
- E2** chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** uses the relationship between algebraic and geometric representations of complex numbers and of conic sections
- E4** uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials
- E5** uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces and resisted motion
- E6** combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions
- E7** uses the techniques of slicing and cylindrical shells to determine volumes
- E8** applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems
- E9** communicates abstract ideas and relationships using appropriate notation and logical argument

Harder applications of the Extension 1 Mathematics course are included in this course. Thus the Outcomes from the Extension 1 Mathematics course are included.

From the Extension 1 Mathematics Course

Preliminary course

- PE1** appreciates the role of mathematics in the solution of practical problems
- PE2** uses multi-step deductive reasoning in a variety of contexts
- PE3** solves problems involving inequalities, polynomials, circle geometry and parametric representations
- PE4** uses the parametric representation together with differentiation to identify geometric properties of parabolas
- PE5** determines derivatives that require the application of more than one rule of differentiation
- PE6** makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations

HSC course

- HE1** appreciates interrelationships between ideas drawn from different areas of mathematics
- HE2** uses inductive reasoning in the construction of proofs
- HE3** uses a variety of strategies to investigate mathematical models of situations involving projectiles, simple harmonic motion or exponential growth and decay
- HE4** uses the relationship between functions, inverse functions and their derivatives
- HE5** applies the chain rule to problems including those involving velocity and acceleration as functions of displacement
- HE6** determines integrals by reduction to a standard form through a given substitution
- HE7** evaluates mathematical solutions to problems and communicates them in an appropriate form

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

QUESTION 1 (15 Marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{1+x}{4+x^2} dx$ 2

(b) By completing the square find $\int \frac{1}{\sqrt{6-x^2-x}} dx$ 2

(c) Find $\int \sin^3 x \cos^3 x dx$ 2

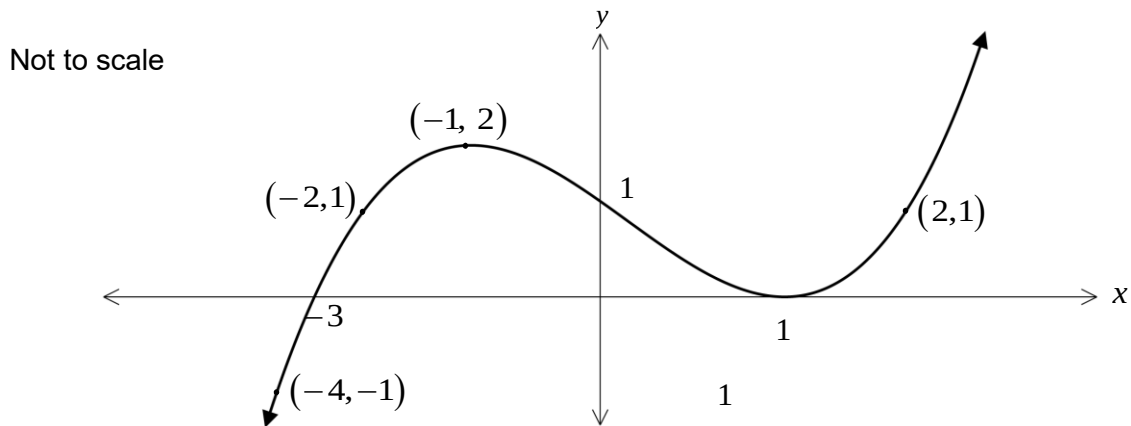
(d) Use integration by parts to evaluate $\int_0^{\ln 2} x e^{-x} dx$. Give your answer in simplest form. 3

(e) By making the numerator rational, or otherwise, find $\int \sqrt{\frac{5-x}{5+x}} dx$ 3

(f) Use a trigonometric substitution to find $\int \frac{dx}{x^2 \sqrt{4-x^2}}$ 3

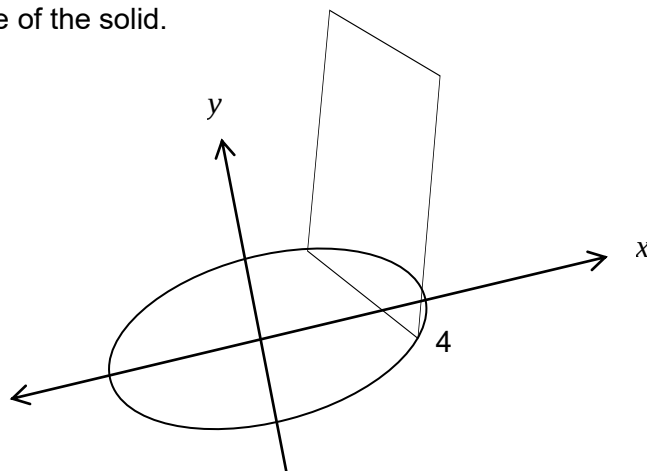
QUESTION 2 (15 Marks) Use a SEPARATE writing booklet.

- (a) The following is a sketch of a function $y = f(x)$



Draw separate one-third page sketches of the following: (clearly showing important features)

- (i) $y = -f(x)$ 1
- (ii) $y = \sqrt{f(x)}$ 1
- (iii) $y = f(1-x)$ 2
- (iv) $y = \cos^{-1} f(x)$ 2
- (v) $y = \frac{1}{1-f(x)}$ 2
- (b) Write down the equation of $P(x)$ if it is a monic polynomial of degree 3 with integer coefficients, a constant term of 12 and one root equal to $\sqrt{3}$. Leave your answer in factored form. 2
- (c) Evaluate $\int_0^3 |x+1| dx$ 2
- (d) The base of a solid is in the circle $x^2 + y^2 = 16$ and every plane section perpendicular to the x axis is a rectangle whose height is twice its base (which lies inside the circle). Find the volume of the solid. 3



QUESTION 3 (15 Marks) Use a SEPARATE writing booklet.

(a) Let $z = \frac{2 - 3i}{1 + i}$

(i) Find $\bar{z}$ in the form $x + iy$

2

(ii) Evaluate $|z|$

1

(b) Consider $w = -\sqrt{3} + i$

(i) Express w in modulus-argument form

2

(ii) Hence or otherwise show that $w^7 + 64w = 0$

2

(c) Sketch the region in the complex plane where the inequalities $1 \leq |z - i| \leq 2$ and $\text{Im}(z) \geq 0$ hold simultaneously.

Clearly mark in all x and y intercepts.

3

(d) In an Argand diagram z is a point on the circle $|z| = 2$.

Given that $\arg z = \theta$ and $0 < \theta < \frac{\pi}{2}$

(i) Draw a diagram to represent this information.

1

(ii) Find, in terms of θ , an expression for $\arg z^2$

1

(iii) Find, in terms of θ , giving brief reasons, expressions for :

(A) $\arg(z + 2)$

1

(B) $\arg(z - 2)$

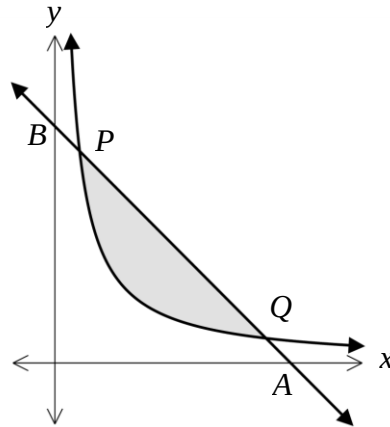
1

(C) $\left| \frac{z - 2}{z + 2} \right|$

1

QUESTION 4 (15 Marks) Use a SEPARATE writing booklet.

(a) Consider the rectangular hyperbola $xy = c^2$ where $c > 0$.



- (i) Prove that the equation of the chord joining points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ where $0 < p < q$ is given by $x + pqy = c(p + q)$. 2

- (ii) The chord PQ intersects the x and y axes at A and B respectively. Prove $AP = BQ$. 2

- (iii) Show that the area enclosed by the hyperbola $xy = c^2$ and chord PQ is $\frac{c^2(q^2 - p^2)}{2pq} + c^2 \ln\left(\frac{p}{q}\right)$ square units. 2

- (b) (i) Divide the polynomial $P(x) = x^4 + 3x^3 - 7x^2 + 11x - 1$ by $x^2 + 2$ and write your result in the form $P(x) = (x^2 + 2)Q(x) + cx + d$. 2

- (ii) Hence determine the values of a and b for which the polynomial $(x^4 + 3x^3 - 7x^2 + 2x) + ax + b$ is exactly divisible by $x^2 + 2$. 2

(c) The equation $|z - 3| + |z + 3| = 10$ corresponds to an ellipse in the Argand diagram.

- (i) Prove that the equation of the ellipse is $\frac{x^2}{25} + \frac{y^2}{16} = 1$ 3

- (ii) Sketch the ellipse showing all important features. 2

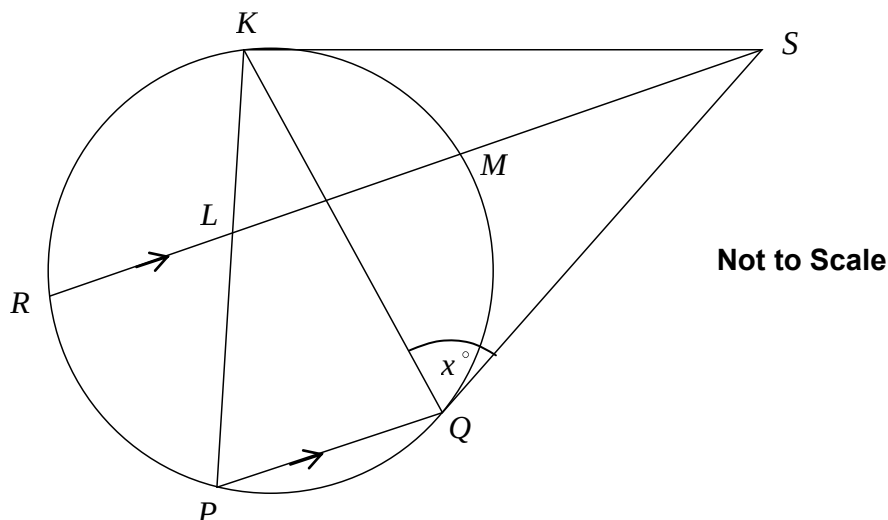
QUESTION 5 (15 Marks) Use a SEPARATE writing booklet.

Marks

- (a) If $u_1 = 1$, $u_2 = 5$ and $u_n = 5u_{n-1} - 6u_{n-2}$ for integers $n \geq 3$, prove by induction that $u_n = 3^n - 2^n$ for integers $n \geq 1$.

3

- (b) In the diagram below PQ and RM are parallel chords in a circle. The tangent at Q meets RM produced at S and SK is another tangent to the circle. PK cuts RM at L .



- (i) Copy or trace this diagram into your answer booklet.
Let $\angle SQK = x^\circ$ and prove $\angle SQK = \angle SLK$

2

- (ii) Explain why $LKSQ$ is a cyclic quadrilateral.

1

- (iii) Prove $PL = QL$

3

- (c) It is given that $\sum_{r=0}^n (-1)^r \frac{{}^nC_r}{x+n} = \frac{n!}{x(x+1)(x+2)\dots(x+n)}$. (DO NOT PROVE)

Hence prove $1 - \frac{1}{2} {}^nC_1 + \frac{1}{3} {}^nC_2 - \dots \frac{(-1)^n {}^nC_n}{n+1} = \frac{1}{n+1}$

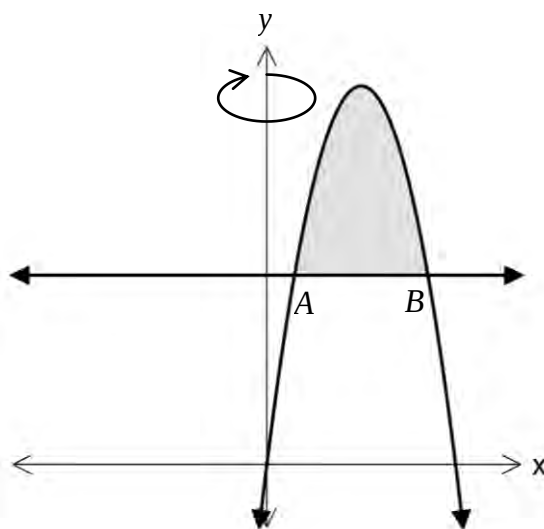
2

Question 5 continues on the next page.

Question 5 (continued)

Marks

- (d) The curve $y = 8x - x^2$ and the line $y = 12$ is sketched below.



- | | | |
|------|--|---|
| (i) | Find the coordinates of the points of intersection A and B | 1 |
| (ii) | <p>The shaded area is rotated around the y axis.</p> <p>Use the method of cylindrical shells to find the exact volume formed. (You may leave your answer unsimplified in fractional form)</p> | 3 |

QUESTION 6 (15 Marks) Use a SEPARATE writing booklet.

(a) (i) If $\frac{1}{x(\pi-2x)} = \frac{A}{x} + \frac{B}{\pi-2x}$ and $A = \frac{1}{\pi}$ find B in terms of π . 1

(ii) Hence show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{x(\pi-2x)} = \frac{2}{\pi} \ln 2$ 3

(iii) By using the substitution $u = a + b - x$ show that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ 1

(iv) Hence evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x dx}{x(\pi-2x)}$ 3

(b) A curve is defined by the equation $y = \frac{a}{2} \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right)$

(i) Show that $1 + \left(\frac{dy}{dx} \right)^2 = \frac{y^2}{a^2}$ 3

(ii) The arc length S between points $(0, a)$ and (x, y) of the curve is given by

$$S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \quad (\text{DO NOT PROVE THIS})$$

Show that $S = \sqrt{y^2 - a^2}$ 4

QUESTION 7 (15 Marks) Use a SEPARATE writing booklet.

- (a) (i) Explain why the domain of the function, $f(x) = \sqrt{2 - \sqrt{x}}$ is $0 \leq x \leq 4$ 1
- (ii) Show that $f(x)$ is a decreasing function and hence find its range. 2
- (iii) Using the substitution, $u = 2 - \sqrt{x}$ or otherwise, find the area bounded by the curve and the x and y axes. 3
- (b) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ where n is an integer and $n \geq 3$.
Show that $I_n + I_{n-2} = \frac{1}{n-1}$ 3
- (c) A body mass of 1 kg falls vertically downwards, from rest, in a medium which exerts a resistance to its motion of $\frac{1}{100} v^2$ Newtons (where v metres per second is the speed of the body when it has fallen a distance of x metres).
(i) Show (on a diagram) that the equation of motion of the body is $\ddot{x} = g - \frac{1}{100} v^2$ where g is the acceleration due to gravity. 1
(ii) Show that the terminal speed V_T is given by $V_T = 10\sqrt{g}$ 2
(iii) Prove that $v^2 = (V_T)^2 \left(1 - e^{-\frac{x}{50}}\right)$ 3

QUESTION 8 (15 Marks) Use a SEPARATE writing booklet.**Marks**

(a) A curve is defined implicitly by the equation $x^2 + 2xy + y^5 = 4$

(i) Show that the gradient of the tangent at $P(X, Y)$ is given by

$$\frac{dy}{dx} = \frac{-2X - 2Y}{5Y^4 + 2X} \quad 2$$

(ii) The tangent is horizontal at P . Show that X satisfies $X^5 + X^2 + 4 = 0$. 1

(iii) Show that X is the unique real solution of $X^5 + X^2 + 4 = 0$ and that $-2 < X < -1$ 3

(b) (i) Solve $\tan 4\theta = 1$ for $0 \leq \theta \leq \pi$ 1

(ii) Express $\tan 2\theta$ in terms of $\tan \theta$. 1

(iii) Hence show $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$. 2

(iv) Hence show $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ has roots $\tan \frac{\pi}{16}$, $\tan \frac{5\pi}{16}$, $\tan \frac{9\pi}{16}$ and $\tan \frac{13\pi}{16}$. 2

(v) Hence evaluate $\tan \frac{\pi}{16} \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} \tan \frac{13\pi}{16}$ 1

(vis1) By solving $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ another way, show the exact value of

$$\tan \frac{\pi}{16} - \cot \frac{\pi}{16} = -2 - 2\sqrt{2}. \quad 2$$

END OF PAPER

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1

$$\begin{aligned} a) \int \frac{1+x}{4+x^2} dx &= \int \frac{1}{4+x^2} dx + \frac{1}{2} \int \frac{2x}{4+x^2} dx \\ &= \frac{1}{2} \tan^{-1} \frac{x}{2} + \frac{1}{2} \ln(4+x^2) + C \end{aligned}$$

$$\begin{aligned} b) \int \frac{1}{\sqrt{6-x^2-x}} dx &= \int \frac{1}{\sqrt{-(x^2+x-6)}} \\ &= \int \frac{1}{\sqrt{-(x^2+x+\frac{1}{4})-\frac{25}{4}}} \\ &= \int \frac{1}{\sqrt{(\frac{x}{2})^2 - (x+\frac{1}{2})^2}} \\ &= \sin^{-1} \frac{x+\frac{1}{2}}{\frac{5}{2}} + C \\ &= \sin^{-1} \frac{2x+1}{5} + C \end{aligned}$$

$$\begin{aligned} c) \int \sin^3 x \cos^3 x dx &= \int \sin^2 x \cos^2 x (1-\sin^2 x) dx \\ &= \int \sin^2 x \cos^2 x - \int \sin^4 x \cos^2 x dx \\ &= \frac{\sin^4 x}{4} - \frac{\sin^6 x}{6} + C \end{aligned}$$

$$\begin{aligned} d) \int_0^{\ln 2} x e^{-x} dx & \quad \text{let } u=x \quad dv=e^{-x} dx \\ & \quad du=dx \quad v=-e^{-x} \\ &= [-x e^{-x}]_0^{\ln 2} + \int_0^{\ln 2} e^{-x} dx \\ &= -\ln 2 e^{-\ln 2} - 0 + [-e^{-x}]_0^{\ln 2} \\ &= -\ln 2 \times \frac{1}{2} + (-e^{-\ln 2} - e^0) \\ &= -\frac{\ln 2}{2} - \frac{1}{2} + 1 = \frac{1-\ln 2}{2} \end{aligned}$$

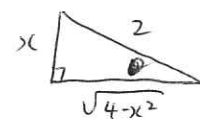
$$Q1 e) \int \frac{\sqrt{5-x}}{\sqrt{5+x}} \cdot \frac{\sqrt{5-x}}{\sqrt{5-x}} = \int \frac{5-x}{\sqrt{25-x^2}} dx$$

$$= \int \frac{5}{\sqrt{25-x^2}} dx + \frac{1}{2} \int \frac{-2x}{\sqrt{25-x^2}} dx$$

$$= 5 \sinh^{-1} \frac{x}{5} + \sqrt{25-x^2} + C$$

$$f) I = \int \frac{dx}{x^2 \sqrt{4-x^2}}$$

construct right angled Δ
with smaller length $\sqrt{4-x^2}$:



$$\Rightarrow \sinh \theta = \frac{x}{2}$$

$$x = 2 \sinh \theta$$

$$dx = 2 \cosh \theta d\theta$$

$$\text{also, } \cos \theta = \frac{\sqrt{4-x^2}}{2}$$

$$\therefore \sqrt{4-x^2} = 2 \cos \theta$$

$$\therefore I = \int \frac{2 \cosh \theta d\theta}{4 \sinh^2 \theta \cdot 2 \cos \theta}$$

$$= \frac{1}{4} \int \frac{d\theta}{\sinh^2 \theta}$$

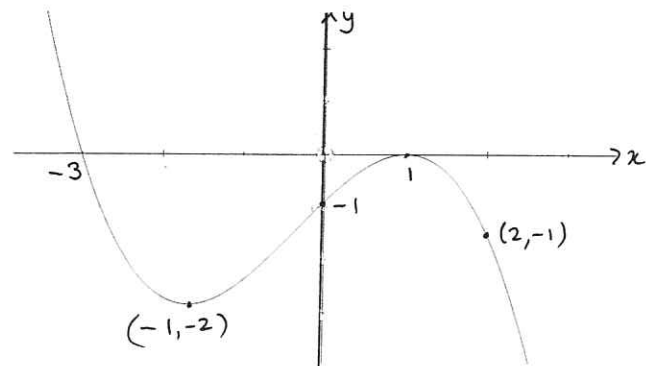
$$= \frac{1}{4} \int \operatorname{cosec}^2 \theta d\theta$$

$$= -\frac{1}{4} \cot \theta + C$$

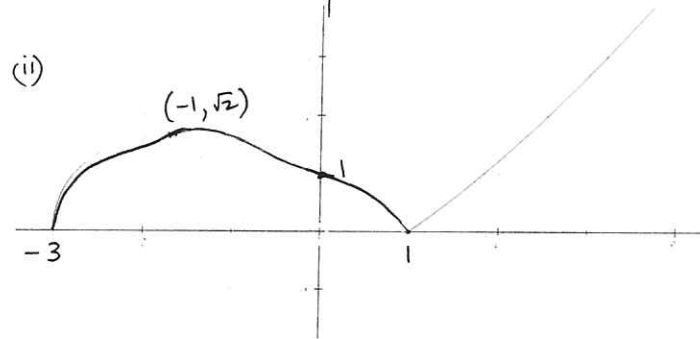
$$= -\frac{\sqrt{4-x^2}}{4x} + C$$

2) (a) (i) All diagrams are not strictly to scale.

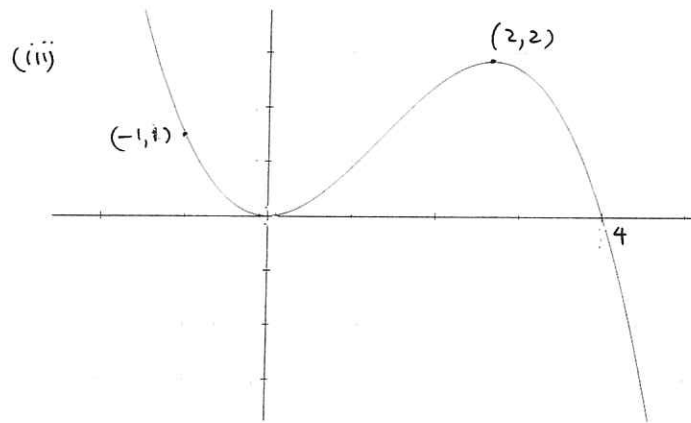
(3)



$$y = -f(x)$$



$$y = \sqrt{f(x)}$$



$$y = f(1-x) = f(-(x-1))$$

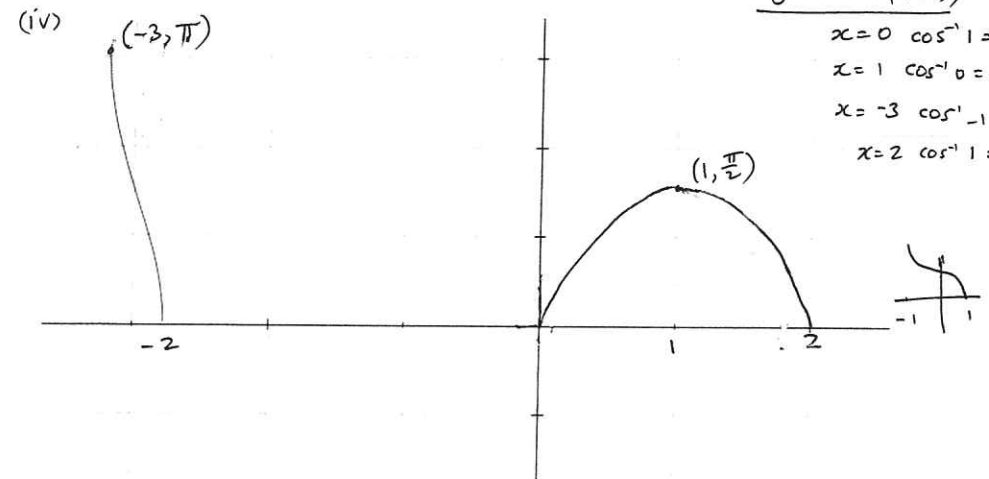
flip over y axis
and shift 1 to
the right

$$\begin{aligned} x=0 & f(1)=0 \\ x=1 & f(0)=1 \\ x=2 & f(-1)=2 \\ x=-1 & f(2)=1 \\ x=3 & f(-2)=0 \\ x=4 & f(-3)=0 \end{aligned}$$

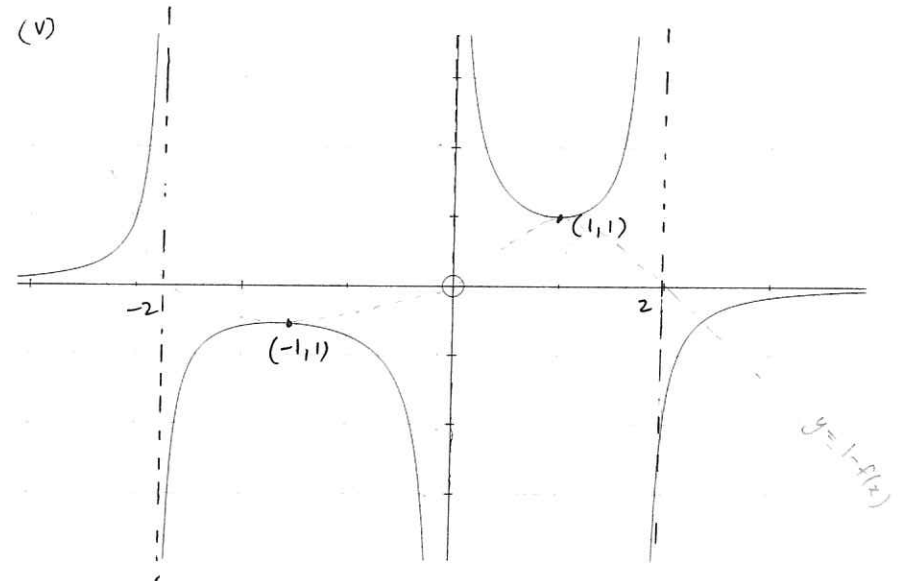
(4)

$$y = \cos^{-1}(f(x))$$

$$\begin{aligned} x=0 & \cos^{-1}1=0 \\ x=1 & \cos^{-1}0=\frac{\pi}{2} \\ x=-3 & \cos^{-1}(-1)=\pi \\ x=2 & \cos^{-1}1=0 \end{aligned}$$



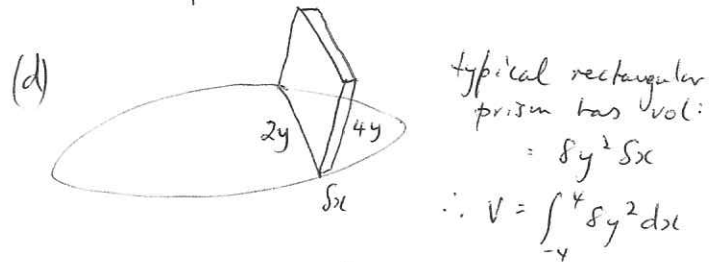
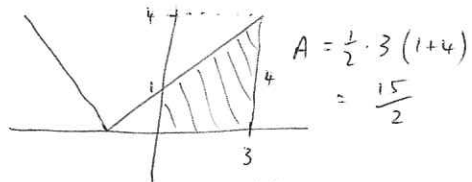
Gap between $x=-2$ and $x=0$ as $f(x) > 1$ [not defined for $\cos^{-1}$]



Sketch $y = 1-f(x)$ (shift (i) up by 1 unit)
Then sketch reciprocal.

Q2b) $P(x) = (x^2 - 3)(x - 4)$

c) $\int_0^3 |x+1| dx = \text{area under graph:}$



$\therefore V = \int_{-4}^4 8y^2 dx$

given the symmetry $V = 2 \int_0^4 8y^2 dx$

$= 16 \int_0^4 16 - x^2 dx$

$= 16 \left[16x - \frac{x^3}{3} \right]_0^4$

$= 16 \left[(64 - \frac{64}{3}) - 0 \right]$

$= \frac{2048}{3} u^3$

(5)

Q3

(a)(i) $\frac{2-3i}{1+i} \cdot \frac{1-i}{1-i}$

$= \frac{2-2i-3i+3i^2}{1+i}$

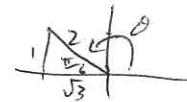
$= -\frac{1}{2} - \frac{5i}{2}$

$\therefore \bar{Z} = -\frac{1}{2} + \frac{5i}{2}$

(ii) $|Z| = \sqrt{\frac{1}{4} + \frac{25}{4}}$
 $= \frac{\sqrt{26}}{2}$

(b)(i) $\omega = -\sqrt{3} + i$

$\therefore \omega = 2 \cos \frac{5\pi}{6}$



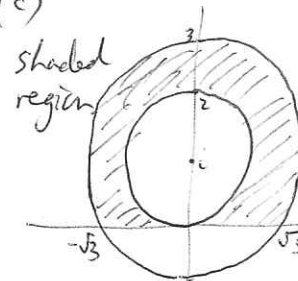
(ii) $\omega^7 + 64\omega = 2^7 \cos \frac{35\pi}{6} + 64 \cdot 2 \cos \frac{5\pi}{6}$

$= 128 \cos \left(-\frac{\pi}{6} \right) + 128 \cos \frac{5\pi}{6}$

$= -128 \cos \frac{5\pi}{6} + 128 \cos \frac{5\pi}{6}$

$= 0$

(c)



For x intercepts, larger circle given by

$x^2 + (y-1)^2 = 4$

when $y=0$

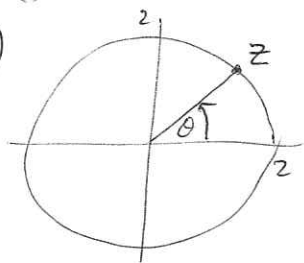
$x^2 + 1 = 4$

$x^2 = 3$

$x = \pm \sqrt{3}$

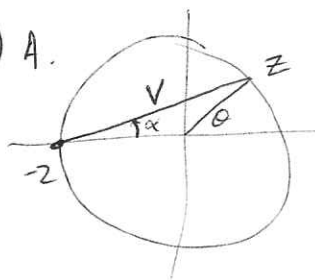
Q3 (d)

(i)



$$(ii) \arg z^2 = 2 \arg z \\ = 2\theta$$

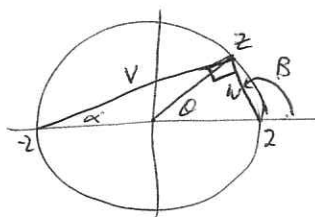
(iii) A.



$z+2$ can be represented by the vector, V .

$$\arg(z+2) = \alpha \\ \text{where } \alpha = \frac{\theta}{2} \text{ (angle at the centre = twice angle at the circumference)}$$

B

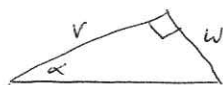


$z-2$ can be represented by the vector, W

$$\arg(z-2) = \beta \\ \beta = \alpha + 90^\circ \text{ (ext angle of } \Delta, \text{ the } \Delta \text{ is right-angled (angle in semi-circle))} \\ = \frac{\theta}{2} + 90$$

$$C. \left| \frac{z-2}{z+2} \right| = \frac{|z-2|}{|z+2|} = \frac{|W|}{|V|}$$

$$\text{Now } \tan \alpha = \frac{|W|}{|V|} \\ \text{i.e. } \tan \frac{\theta}{2} = \frac{|W|}{|V|}$$



(7)

Q4

$$a) xy = c^2$$

$$(i) M_{PQ} = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} = \frac{c \frac{q-p}{pq}}{c(p-q)} = \frac{-(p-q)}{\frac{pq}{(p-q)}} = -\frac{1}{pq}$$

$$\therefore y - \frac{c}{p} = -\frac{1}{pq} (x - cp)$$

$$pqy - cq = -x + cp$$

$$x + pqy = c(p+q)$$

$$(ii) \text{ Chord at } x \text{ axis: } x = c(p+q) \text{ i.e. } A[c(p+q), 0]$$

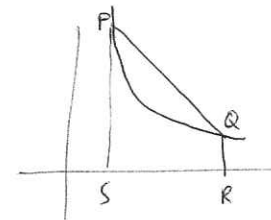
$$\text{at } y \text{ axis: } pqy = c(p+q) \text{ i.e. } B(0, \frac{c(p+q)}{pq})$$

$$(d_{AP})^2 = (cp - c(p+q))^2 + (\frac{c}{p})^2 \\ = c^2 q^2 + \frac{c^2}{p^2}$$

$$(d_{BQ})^2 = (cq - 0)^2 + [\frac{c}{q} - \frac{c(p+q)}{pq}]^2 \\ = c^2 q^2 + [\frac{cp - cp - cq}{pq}]^2 \\ = c^2 q^2 + \frac{c^2}{p^2} \\ = (d_{AP})^2$$

$$\therefore d_{AP} = d_{BQ}$$

$$(iii) \text{ Area} = \text{trapezium PQRS} - \int_{cp}^{cq} \frac{c^2}{x} dx \\ = \frac{1}{2} c(q-p) [c(\frac{1}{p} + \frac{1}{q})] - c^2 \int_{cp}^{cq} \frac{1}{x} dx \\ = \frac{1}{2} c^2 (q-p) (\frac{q+p}{pq}) - c^2 [\ln x]_{cp}^{cq} \\ = \frac{1}{2} c^2 (\frac{q^2 - p^2}{pq}) - c^2 (\ln cq - \ln cp) \\ = \frac{c^2 (q^2 - p^2)}{2pq} + c^2 \ln \left(\frac{p}{q} \right)$$



$$\ln cq - \ln cp \\ = \ln \frac{cq}{cp} \\ = -\ln \frac{p}{q}$$

(8)

4(b) (i)
$$\begin{array}{r} x^2 + 3x - 9 \\ x^2 + 2 \overline{) x^4 + 3x^3 - 7x^2 + 11x - 1} \\ \underline{-(x^4 + + 2x^2)} \\ 3x^3 - 9x^2 \\ \underline{-(3x^3 + 6x)} \\ -9x^2 + 5x \\ \underline{-(-9x^2 - 18)} \\ 5x + 17 \end{array}$$

$\therefore P(x) = (x^2 + 2)[x^2 + 3x - 9] + 5x + 17$

(ii) $x^2 + 2$ divides into the first 3 terms as above, but the algorithm gets modified with the $2x$:

$$\begin{array}{r} 2x \\ -(+ 6x) \\ \hline -9x^2 - 4x \\ -(-9x^2 - 18) \\ \hline -4x + 18 \leftarrow \text{Remainder} \end{array}$$

$\therefore ax + b \equiv 4x - 18$ to ensure the remainder cancels.

alternatively: use result

$$x^4 + 3x^3 - 7x^2 + 11x - 1 = (x^2 + 2)(x^2 + 3x - 9) + 5x + 17$$

$$x^4 + 3x^3 - 7x^2 + 11x - 1 - 5x - 17 = (x^2 + 2)(x^2 + 3x - 9)$$

$$x^4 + 3x^3 - 7x^2 + 6x - 18 = (x^2 + 2)(x^2 + 3x - 9)$$

$$b = -18$$

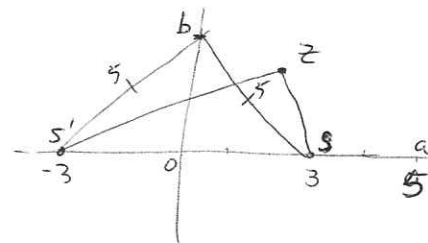
$$2x + ax = 6x$$

$$a = 4$$

9

4(c)

(i) $|z - 3| + |z + 3| = 10$



$$zS + zS' = 2a \text{ by definition.} \\ = 10$$

$$\therefore a = 5$$

$$b^2 = 25 - 9 \text{ (from } \triangle OSb) \\ = 16$$

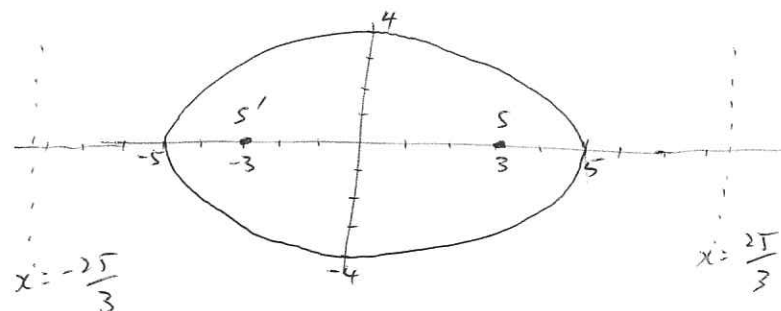
$$\therefore b = 4$$

$$\text{Hence } \frac{x^2}{25} + \frac{y^2}{16} = 1$$

(ii) Directrices $x = \pm \frac{a}{e}$ ($e = \frac{3}{5}$)

$$\therefore x = \pm 5 \times \frac{5}{3} \\ = \pm \frac{25}{3}$$

$$\text{foci } (\pm 3, 0)$$



Q 5

$$(a) u_1 = 1 \quad u_2 = 5 \quad u_n = 5u_{n-1} - 6u_{n-2}$$

$$\text{Prove } u_n = 3^n - 2^n$$

$$\begin{aligned} \text{For } n=3: \quad u_3 &= 5u_2 - 6u_1 \\ &= 5(5) - 6(1) \\ &= 19 \end{aligned}$$

1. Show true for $n=3$

$$\begin{aligned} u_3 &= 3^3 - 2^3 \\ &= 27 - 8 = 19 \end{aligned}$$

$\therefore$ true for $n=3$

2. Assume true for $n=k$

$$\text{i.e. given } u_k = 5u_{k-1} - 6u_{k-2}$$

$$\text{then } u_k = 3^k - 2^k$$

3. Prove true for $n=k+1$

$$\text{i.e. given } u_{k+1} = 5u_k - 6u_{k-1}$$

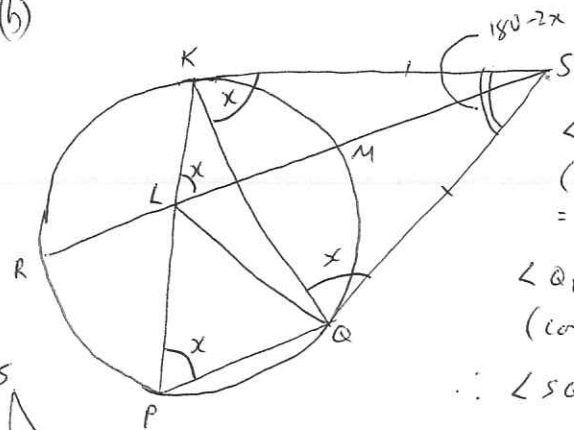
$$\text{show that } u_{k+1} = 3^{k+1} - 2^{k+1}$$

$$\begin{aligned} \text{LHS} &= 5u_k - 6u_{k-1} \\ &= 5(3^k - 2^k) - 6(3^{k-1} - 2^{k-1}) \\ &= 5(3^k - 2^k) - [2 \cdot 3 \cdot 3^{k-1}] + [3 \cdot 2 \cdot 2^{k-1}] \\ &= 5 \cdot 3^k - 5 \cdot 2^k - 2 \cdot 3^k + 3 \cdot 2^k \\ &= 3 \cdot 3^k - 2 \cdot 2^k \\ &= 3^{k+1} - 2^{k+1} \\ &= \text{RHS} \end{aligned}$$

(11)

Q 5(b)

(i)

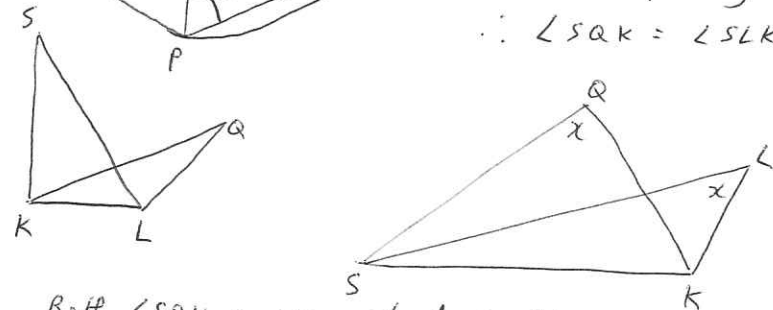


$\angle SQR = \angle QPK = x$
($\angle$ between tangent & chord
= $\angle$ in alt segment).

$\angle QPK = \angle SLK$
(corresponding $\angle$'s in $\parallel$ lines =)

$$\therefore \angle SQR = \angle SLK$$

(ii)



Both $\angle SQR$ & $\angle SLK$ stand on same arc KS
of circle $LKSQ$

(iii) $\triangle KLS$ is isosceles with $KS = LS \therefore \angle SKL = \angle SLK = x$
in $\triangle KLS$, $\angle KLS = 180 - 2x$ ($\angle$ sum of $\triangle$)

$\therefore$ in cyclic quad ($\angle KLS$) $\angle KLS = 2x$ (suppl with $\angle KLS$).

now $\angle KLS$ is ext $\angle$ of $\triangle PLS$.

$$\therefore \angle PLS = x$$

$\therefore \triangle PLS$ is isosceles

$\therefore \angle L = \angle P$ (sides opp = $\angle$'s).

(12)

5(c)

$$\sum_{r=0}^n (-1)^r \frac{n C_r}{x+n} = \frac{n!}{x(x+1)(x+2) \dots (x+n)}$$

if $x=1$:

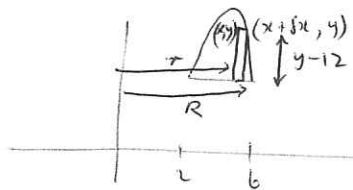
$$\sum_{r=0}^n (-1)^r \frac{n C_r}{1+n} = \frac{n!}{1(2)(3) \dots (n+1)}$$

$$\begin{aligned} \text{RHS} &= \frac{n!}{(n+1)!} \\ &= \frac{1}{n+1} \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \frac{(-1)^0 n C_0}{1} + \frac{(-1)^1 n C_1}{2} + \frac{(-1)^2 n C_2}{3} + \dots + \frac{(-1)^n n C_n}{1+n} \\ &= 1 - \frac{n C_1}{2} + \frac{n C_2}{3} - \dots + \frac{(-1)^n}{1+n} \end{aligned}$$

$$\therefore 1 - \frac{1}{2} n C_1 + \frac{1}{3} n C_2 - \dots + \frac{(-1)^n}{1+n} = \frac{1}{n+1}$$

(d) $12 = 8x - x^2$
 $x^2 - 8x + 12 = 0$
 $(x-6)(x-2) = 0$
 POI: $(2, 12)$ & $(6, 12)$



$$\begin{aligned} \text{X-sect Area of shell} &= \pi(R^2 - r^2) \\ &= \pi((x+dx)^2 - x^2) \\ &= \pi(x^2 + 2x dx + dx^2 - x^2) \\ &= \pi 2x dx \quad (\text{ignoring } dx^2) \\ &= 2\pi x dx \end{aligned}$$

$$\begin{aligned} \text{Vol of typical shell} \\ dV &= 2\pi x dx (y-12) \end{aligned}$$

 $\therefore$ total volume

$$= 2\pi \int_2^6 x(8x - x^2 - 12) dx$$

$$= 2\pi \int_2^6 8x^2 - x^3 - 12x dx$$

$$= 2\pi \left[\frac{8x^3}{3} - \frac{x^4}{4} - 6x^2 \right]_2^6$$

$$= 2\pi \left[\left(\frac{8 \times 6^3}{3} - \frac{6^4}{4} - 6 \times 36 \right) - \left(\frac{8 \times 2^3}{3} - \frac{2^4}{4} - 6 \times 4 \right) \right]$$

(13)

Q6

$$a(i) \quad A \frac{(\pi - 2x) + Bx}{x(\pi - 2x)} = \frac{1}{x(\pi - 2x)}$$

$$\frac{1}{\pi} (\pi - 2x) + Bx = 1$$

$$1 - \frac{2x}{\pi} + Bx = 1$$

$$Bx = \frac{2x}{\pi}$$

$$\therefore B = \frac{2}{\pi}$$

$$(ii) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\pi} \frac{1}{x} + \frac{1}{\pi} \left(\frac{2}{\pi - 2x} \right)$$

$$= \frac{1}{\pi} \ln x - \frac{1}{\pi} \ln (\pi - 2x)$$

$$= \frac{1}{\pi} [\ln x - \ln (\pi - 2x)]$$

$$= \frac{1}{\pi} \left[\ln \frac{x}{\pi - 2x} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{1}{\pi} \left[\ln \frac{\frac{\pi}{3}}{\frac{\pi}{3}} - \ln \frac{\frac{\pi}{6}}{\frac{\pi}{6}} \right]$$

$$= \frac{1}{\pi} [0 - \ln \frac{1}{4}]$$

$$= \frac{1}{\pi} [-\ln 2^{-2}]$$

$$= \frac{1}{\pi} [2 \ln 2]$$

$$= \frac{2}{\pi} \ln 2$$

(14)

$$(iii) \int_a^b f(x) dx$$

$$\text{Let } u = a+b-x \Rightarrow du = -dx$$

$$x = a+b-u$$

$$\therefore dx = -du$$

$$\text{when } x=a, u=b$$

$$x=b, u=a$$

$$\begin{aligned} \therefore \int_a^b f(x) dx &= \int_b^a f(a+b-u) (-du) \\ &= \int_a^b f(a+b-u) du \\ &= \int_a^b f(a+b-x) dx \end{aligned}$$

$$(iv) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x dx}{x(\pi-2x)} = \int \frac{\cos^2 \left[\frac{\pi}{6} + \frac{\pi}{3} - x \right] dx}{\left(\frac{\pi}{6} + \frac{\pi}{3} - x \right) \left[\pi - 2 \left(\frac{\pi}{6} + \frac{\pi}{3} - x \right) \right]}$$

$$= \int \frac{\cos^2 \left(\frac{\pi}{2} - x \right) dx}{\left(\frac{\pi}{2} - x \right) \left[\pi - 2 \left(\frac{\pi}{2} - x \right) \right]}$$

$$= \int \frac{\sin^2 x}{\left(\frac{\pi}{2} - x \right) (2x)} dx$$

$$= \int \frac{1 - \cos^2 x}{x(\pi - 2x)} dx$$

$$\text{So, } \int \frac{\cos^2 x dx}{x(\pi-2x)} = \int \frac{1}{x(\pi-2x)} - \frac{\cos^2 x}{x(\pi-2x)} dx$$

$$\therefore 2 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x dx}{x(\pi-2x)} = \frac{2}{\pi} \ln 2$$

$$\therefore \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x dx}{x(\pi-2x)} = \frac{1}{\pi} \ln 2$$

(15)

$$6(b)(i) \quad y = \frac{a}{2} \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right) \quad \text{and} \quad y^2 = \frac{a^2}{4} \left[e^{\frac{2x}{a}} + 2 + e^{-\frac{2x}{a}} \right] \quad (16)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{a}{2} \left[\frac{1}{a} e^{\frac{x}{a}} - \frac{1}{a} e^{-\frac{x}{a}} \right] \\ &= \frac{1}{2} \left[e^{\frac{x}{a}} - e^{-\frac{x}{a}} \right] \end{aligned}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{1}{4} \left[e^{\frac{2x}{a}} - 2 + e^{-\frac{2x}{a}} \right]$$

$$1 + \left(\frac{dy}{dx} \right)^2 = 1 + \frac{e^{\frac{2x}{a}}}{4} - \frac{1}{2} + \frac{e^{-\frac{2x}{a}}}{4}$$

$$= \frac{1}{4} \left[e^{\frac{2x}{a}} + 2 + e^{-\frac{2x}{a}} \right]$$

$$= \frac{y^2}{a^2}$$

$$(ii) \quad S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx = \int_0^x \sqrt{\frac{y^2}{a^2}} dx$$

$$= \int_0^x \frac{y}{a} dx$$

$$= \frac{1}{2} \int_0^x \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right) dx$$

$$= \frac{1}{2} \left[a e^{\frac{x}{a}} - a e^{-\frac{x}{a}} \right]_0^x$$

$$= \frac{a}{2} \left[e^{\frac{x}{a}} - e^{-\frac{x}{a}} \right] - \frac{a}{2} [a - a]$$

$$\therefore S = \frac{a}{2} \left[e^{\frac{x}{a}} - e^{-\frac{x}{a}} \right]$$

$$\text{And } y^2 - a^2 = \frac{a^2}{4} \left[e^{\frac{2x}{a}} + e^{-\frac{2x}{a}} \right] - a^2$$

$$= \frac{a^2}{4} \left[e^{\frac{2x}{a}} + 2 + e^{-\frac{2x}{a}} - 4 \right]$$

$$= \frac{a^2}{4} \left[e^{\frac{2x}{a}} - 2 + e^{-\frac{2x}{a}} \right]$$

$$= \frac{a^2}{4} \left(e^{\frac{x}{a}} - e^{-\frac{x}{a}} \right)^2$$

$$= S^2$$

$$\therefore S = \sqrt{y^2 - a^2}$$

Q7

$$a(i) f(x) = \sqrt{2-\sqrt{x}} \quad x \geq 0 \text{ for } \sqrt{x} \\ x \leq 4 \text{ for } \sqrt{2-\sqrt{4}}$$

$$(ii) f(x) = (2-x^{\frac{1}{2}})^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2} \cdot (-\frac{1}{2}x^{-\frac{1}{2}})(2-x^{\frac{1}{2}})^{-\frac{1}{2}}$$

$$= \frac{-1}{4\sqrt{x}\sqrt{2-\sqrt{x}}} < 0 \text{ for } 0 \leq x \leq 4$$

$\therefore$ a decreasing function.

$$(iii) u = 2 - \sqrt{x}$$

$$du = -\frac{1}{2}x^{-\frac{1}{2}}dx$$

$$= -\frac{1}{2\sqrt{x}}dx$$

$$\therefore dx = -2\sqrt{x}du$$

$$= -2(2-u)du$$

$$\text{when } x=4, u=0$$

$$x=0, u=2$$

$$x=0 \quad f(x)=\sqrt{2}$$

$$x=4 \quad f(x)=0$$

$$\text{Range } 0 \leq y \leq \sqrt{2}$$

$$\therefore \int_0^4 \sqrt{2-\sqrt{x}} dx$$

$$= -2 \int_2^0 \sqrt{u}(2-u)du$$

$$= 2 \int_0^2 2u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$

$$= 2 \left[\frac{4u^{\frac{3}{2}}}{3} - \frac{2u^{\frac{5}{2}}}{5} \right]_0^2$$

$$= 2 \left[\left(\frac{4}{3} 2^{\frac{3}{2}} - \frac{2}{5} 2^{\frac{5}{2}} \right) - (0-0) \right]$$

$$= 2 \left(\frac{8}{3} \sqrt{2} - \frac{8}{5} \sqrt{2} \right) = \frac{32\sqrt{2}}{15}$$

$$7(b) I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^2 x \tan^{n-2} x dx$$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \tan^{n-2} x dx$$

$$= \int_0^{\frac{\pi}{4}} \sec^2 x \tan^{n-2} x dx - \int_0^{\frac{\pi}{4}} \tan^{n-2} x dx$$

$$= \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$I_n = \left(\frac{1}{n-1} - 0 \right) - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

(17)

$$(c)(i) \begin{array}{c} \uparrow \frac{1}{100}v^2 \\ mg \\ \downarrow \end{array} \quad F = m\ddot{x} = mg - \frac{1}{100}v^2 \quad m=1 \\ \therefore \ddot{x} = g - \frac{1}{100}v^2$$

$$(ii) v_T \text{ occurs when } \ddot{x} = 0$$

$$\therefore 0 = g - \frac{v_T^2}{100}$$

$$v_T^2 = 100g$$

$$v_T = 10\sqrt{g}$$

$$(iii) \ddot{x} = g - \frac{v^2}{100} \quad \text{we want } v = f(x)$$

$$v \frac{dv}{dx} = g - \frac{v^2}{100}$$

$$= \frac{100g - v^2}{100}$$

$$\therefore \frac{dv}{dx} = \frac{100g - v^2}{100v} \Rightarrow \frac{dx}{dv} = \frac{100v}{100g - v^2}$$

$$x = \int_0^v \frac{100v}{100g - v^2} dv$$

$$x = -50 \int_0^v \frac{-2v}{100g - v^2} dv$$

$$= [-50 \ln |100g - v^2|]_0^v$$

$$\therefore x = -50 \ln(100g - v^2) + 50 \ln(100g) \quad \text{now write } v = f(x)$$

$$= 50 \ln \frac{100g}{100g - v^2}$$

$$\frac{x}{50} = \ln \frac{100g}{100g - v^2}$$

$$e^{\frac{x}{50}} = \frac{100g}{100g - v^2}$$

$$\frac{100g - v^2}{100g} = e^{-\frac{x}{50}}$$

$$v^2 = 100g - 100g e^{-\frac{x}{50}}$$

$$= 100g(1 - e^{-\frac{x}{50}})$$

$$= v_T^2(1 - e^{-\frac{x}{50}})$$

$$8 (a)(i) x^2 + 2xy + y^2 = 4$$

$$2x + 2y + \frac{dy}{dx} 2x + 5y + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} [2x + 5y] = -2x - 2y$$

$$\therefore \frac{dy}{dx} = \frac{-2x - 2y}{2x + 5y}$$

$$\therefore \omega P(X, Y), \frac{dy}{dx} = \frac{-2X - 2Y}{2X + 5Y}$$

(18)

(8)

(b) (i) $\tan 4\theta = 1 \quad 0 \leq \theta \leq \pi$
 $0 \leq 4\theta \leq 4\pi$

$$4\theta = \frac{\pi}{4}, \frac{\pi}{4} + \pi, 2\pi + \frac{\pi}{4}, 3\pi + \frac{\pi}{4}$$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$

(ii) $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

(iii) Let $x = \tan \theta \quad \tan 4\theta = \frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$
 $= 2 \left(\frac{2t}{1-t^2} \right) \times \frac{(1-t^2)^2}{(1-t^2)^2}$
 $= \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2} = \frac{4t - 4t^3}{1 - 2t^2 + t^4 - 4t^2}$

$$\therefore \tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(iv) Let $\tan 4\theta = 1$ and $\tan \theta = x$

$$1 = \frac{4x - 4x^3}{1 - 6x^2 + x^4}$$

$$x^4 - 6x^2 + 1 = 4x - 4x^3$$

$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

Solns we $x = \tan \theta$ where θ is soln to $\tan 4\theta = 1$

$\therefore$ solns are $\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$

(v) Product of roots = $+\frac{c}{a}$ in $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$
 $= 1$

(vi) $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$

$$x^2 \left(x^2 + 4x - 6 - \frac{4}{x} + \frac{1}{x^2} \right) = 0$$

$$x^2 + 0 \left(\left(x^2 + \frac{1}{x^2} \right) + 4 \left(x - \frac{1}{x} \right) - 6 \right) = 0$$

$$\left(\left(x - \frac{1}{x} \right)^2 + 2 + 4 \left(x - \frac{1}{x} \right) - 6 \right) = 0$$

$$\left(\left(x - \frac{1}{x} \right)^2 + 4 \left(x - \frac{1}{x} \right) - 4 \right) = 0$$

Let $u = x - \frac{1}{x} \quad u^2 + 4u - 4 = 0$
 $u = \frac{-4 \pm \sqrt{16 + 4 \times 4}}{2} = \frac{-4 \pm 4\sqrt{2}}{2} = -2 \pm 2\sqrt{2}$

$$x - \frac{1}{x} = -2 \pm 2\sqrt{2}$$

smallest value is $\tan \frac{\pi}{16} - \frac{1}{\tan \frac{\pi}{16}} = -2 - 2\sqrt{2}$

$$\therefore \tan \frac{\pi}{16} - \cot \frac{\pi}{16} = -2 - 2\sqrt{2}$$

(ii) horiz @ P suggests $-2x - 2y = 0$

$$x = -y$$

$$\therefore x^2 + 2x(-x) + (-x)^5 = 4$$

$$x^2 - 2x^2 - x^5 - 4 = 0$$

$$\therefore x^5 + x^2 + 4 = 0$$

(iii) Let $y = x^5 + x^2 + 4$

$$\frac{dy}{dx} = 5x^4 + 2x$$

Let $\frac{dy}{dx} = 0 \quad 2x + 5x^4 = 0$

$$x(2 + 5x^3) = 0$$

$$x = 0 \text{ or } x^3 = -\frac{2}{5}$$

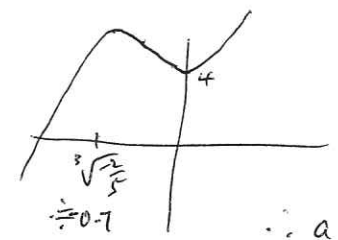
$$x = \sqrt[3]{-\frac{2}{5}}$$

$$\frac{d^2y}{dx^2} = 20x^3 + 2$$

@ $x = 0 \quad \frac{d^2y}{dx^2} = 2 > 0 \therefore U \text{ min.}$

at $x = \sqrt[3]{-\frac{2}{5}} \quad \frac{d^2y}{dx^2} = 20 \left(\sqrt[3]{-\frac{2}{5}} \right)^3 + 2 < 0 \wedge \text{max}$

$\therefore$ min at $(0, 4)$ + max at $x = \sqrt[3]{-\frac{2}{5}}$



When $x = -1, y = -1 + 1 + 4 > 0$

$x = -2, y = -32 + 4 + 4 < 0$

$\therefore$ a unique real solution occurs between -1 & -2 .